WEEK 01 · ASSIGNMENTS

Week 01 assignments

What you are working toward

Task

A clear one-batch planning note

Purpose

Calculate and explain a planning estimate with clear units.

Working material

Week_01.ipynb · Independent lab: one batch planning note

Use the supplied starter and data; implement the marked student tasks.

Task details

Part	What you need to do
Inputs	Bread packs: 2 portions/container; counts 1, 3 and 0.
Calculation	Use named values and multiplication; avoid stored answers.
Output	Show the declaration, container count and planned portions.
Judgment	Explain whether zero is acceptable for an editable draft.

Keep the input contract, units and expected behavior explicit.

A small example of the expected reasoning

Example element	Illustration
Separate example input	3 portions/container × 2 containers
Prediction	6 planned portions
Useful output	2 containers; 6 planned portions
Meaning limit	This does not establish collection, delivery or people served.

Use this reasoning pattern with your own task inputs.

Checks your work should demonstrate

Check	What to look for
Change count	Recalculate after changing an input.
Change description	The label changes; the arithmetic need not change.
Zero draft	State the planning assumption explicitly.
Saved file	Save and rerun a copy of first_pickup.py.

Write expectations before running; explain the first difference.

What to keep as evidence

Evidence	Expected contents
Working artifact	Runnable one-batch note and saved-file practice.
Checks	Predictions and observations for 1, 3 and 0 containers.
Peer revision	One clarified label after another reader checks it.
Explanation	One unresolved question for a coordinator.

A successful run is only part of the evidence; explain your choices.

Research 1: what to investigate

Part	Required evidence
Learning goal	One skill you want to gain and why it matters
Real system	Users, information, action and one failure risk
AI era	One responsibility engineers still hold when AI helps
Reasoning	A connection among your goal, the system and the responsibility

Week 1 → share Week 2 · proposed 5%

Research 1: a short illustrative example

Possible angle

A library loan system: users, book records, borrow/return actions.

Engineering judgment

Prevent the same copy being loaned twice; verify a suggested rule.

Your investigation

Choose a real documented system.
Find a source and explain your own learning goal.

Add a source, your reasoning and a limitation; this example is not a full submission.

Research evidence and submission

Include

Question and sources
Finding in your own words
Concrete example and reasoning
One limitation or question

AI is optional

If used: useful discussion evidence and verified sources.
Explain your own conclusion.

Standard credit

Proposed 5% per research task.
Presentation is not required.
About 15–20 minutes preparation.

No paid tool is required. Full marks are available without AI.

Save, check and submit through the announced route

Save

Keep the working notebook/files and a separate backup.

Check

Open from a clean start. Retain expected and observed results.

Submission

Private route, exact naming and deadlines will be announced.
Keep complete work out of Issues.

Typical work outside class: 30–45 minutes. This guide adds no new grading component.